

Title: When to drop “that”: Revisiting the UID hypothesis on complementizer-dropping in English

Various language production theories have been proposed to account for how speakers choose among alternative ways to express the same message. A paradigmatic example is the syntactic reduction of the complementizer “that” in English [1]. Among these theories, the Uniform Information Density (UID) hypothesis holds that speakers tend to have an even rate of information when uttering a sentence [2,3]. Accordingly, “that” is predicted to be omitted more often when the information rate is low [3]. To evaluate this prediction, researchers have operationalized the information rate at the onset of the complement clause (CC) as the syntactic property of the verb for taking a CC, either with or without the overt “that”, $P(CC|V_k)$ [3], or as the local informativity of the first word in the CC, $P(w_1|context, V_k)$ estimated by language models [4]. However, the local informativity might not be the whole story. In fact, results on phonological reduction suggest that the expected informativity of the target segment is a better predictor than local informativity in predicting whether it will be reduced [5]. The reasoning is that the expected informativity need not be computed online, as it tracks the overall use of the segment across contexts. Hence, if V_k is generally followed by predictable embedded content without an overt “that,” speakers may still omit “that” even when the current CC is surprising. A stronger version of this prediction aligns with the grammaticalization accounts, which posit that omission of “that” has become grammaticalized for certain verbs, especially epistemic markers like *think* and *know* [6]. In the current study, we ask: Would the contextual or expected information content of the verb (V_k) or the first word in CC (w_1) better predict the omission of “that”?

Methods: We randomly selected 1,000,000 sentences from the Dolma v1_6-sample dataset [7] and used Spacy [8] to dependency parse these sentences. This results in 69,573 sentences with a CC, headed by 127 clause-embedding predicates of interest, and 24,938 of these sentences contain an overt complementizer (35.8%). We included non-contextual informativity, local informativity, and expected informativity of the verb (V_k), the syntactic property of the verb taking a CC, and the first word of the CC (w_1) as the information-theoretical measures (Table 1). We used GPT-2 to obtain the local informativity of V_k and w_1 , and estimated the expected informativity using the Monte Carlo method. Since the local and expected informativity of the syntactic information of the CC cannot be directly obtained from the language model, these two measures were left out in the current analysis, but we propose estimation techniques for future work. In addition, we also included control variables to account for the availability effects [1, 3]. We fit a Bayesian logistic mixed-effects model predicting the presence of “that” from the control variables, the main effects of the information-theoretical measures, and the by-verb random intercept. All continuous variables were mean-centered and scaled. The expected informativity of V_k was residualized against the non-contextual informativity of V_k due to their high correlation.

Results: Figure 1 shows the estimated effects of each information-theoretic measure. The non-contextual informativity of V_k and w_1 negatively predicted the omission of “that,” suggesting that the higher surprisal values were linked to a lower likelihood of dropping “that.” In addition, the local informativity of w_1 showed a significant negative effect, consistent with the UID hypothesis [3]. The local informativity of V_k was also a significant predictor, where verbs with higher local informativity were associated with a greater likelihood of “that” omission. By contrast, the residualized expected informativity of the verb did not show a significant effect.

Discussion: The results support the UID hypothesis, suggesting that low local informativity of the verb and the first word in CC strongly predict the speaker’s omission of the optional “that.” The expected informativity of the first word in CC, although significant, shows the opposite effect. Unlike phonological reduction, the effect of the expected informativity of the verb was not significant. Yet, some limitations of the current analysis need to be addressed in the future: First, we plan to include the local informativity and the expected informativity of the syntactic information of the CC. In addition, there might be by-verb variances not captured by the current regression model, such that the effects of the local and expected informativity of the first word of the CC affect the omission of “that” for highly frequent or semantically bleached verbs.

	Verb (V_k)	Complement clause (CC)	First word in CC (w_1)
Non-contextual informativity	$I(V_k) = -\log(\frac{\#V_k}{\sum_i \#V_i})$	$I(CC, V_k) = -\log(\frac{\#CC, V_k}{\#V_k})$	$I(w_1) = -\log(\frac{\#w_1}{\sum_i \#w_i})$
Local informativity	$I(V_k context_k)$ $= -\log P(V_k context_k)$	$I(CC context_k, V_k)$ $= -\log P(CC context_k, V_k)$	$I(w_1 context_k, V_k)$ $= -\log P(w_1 context_k, V_k)$
Expected informativity	$\mathbb{E}[I(context_k V_k)]$ $\approx \frac{1}{\#V_k} \sum_{i: V_i = V_k} I(V_i context_i)$	$\mathbb{E}[I(CC context_k, V_k)]$ $\approx \frac{1}{\#V_k} \sum_{i: V_i = V_k} I(CC context_i, V_i)$	$\mathbb{E}[I(w_1 context_k, V_k)]$ $\approx \frac{1}{\#V_k} \sum_{i: V_i = V_k} I(w_1 context_i, V_i)$

Table 1. Information-theoretical measures that predict complementizer-dropping. Shaded cells represent measures that are not included in the current analysis.

Regression model: complementizer $\sim$ matrix subject type + position of the verb + distance between verb and CC + length of CC onset + length of the remaining CC + **noncontext_verb** + **noncontext_cc** + **noncontext_embedded_n** + **local_verb** + **local_embedded_n** + **expected_verb** + **expected_embedded_n** + (1|verb)

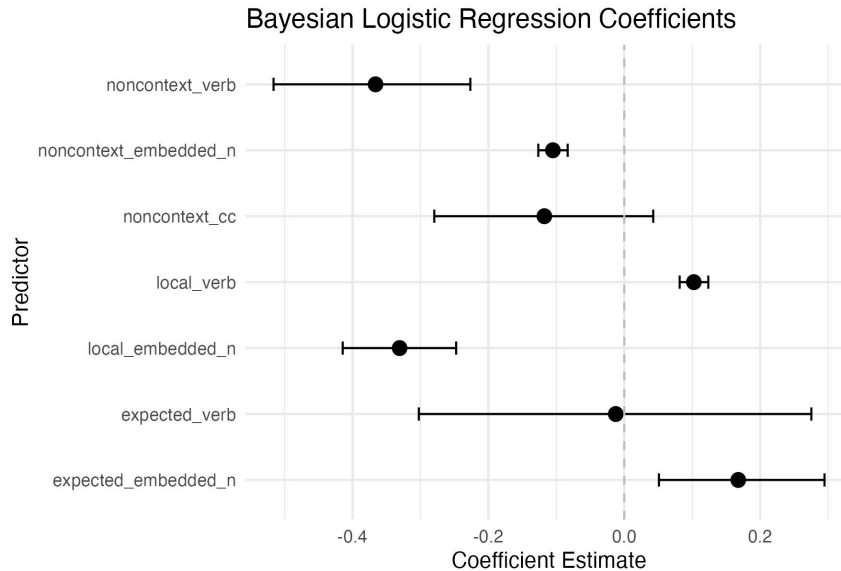


Figure 1. The estimated effect of each information-theoretic measure included in the current Bayesian logistic mixed-effects model. Each dot represents the value of the coefficient, and the error bar shows the 95% credible interval.

References: [1] Ferreira & Dell (2000). *Effect of Ambiguity and Lexical Availability on Syntactic and Lexical Production*. [2] Levy & Jaeger (2006). *Speakers optimize information density through syntactic reduction*. [3] Jaeger (2010). *Redundancy and reduction: Speakers manage syntactic information density*. [4] Rabinovich (2024). *That's Optional: A Contemporary Exploration of "that" Omission in English Subordinate Clauses*. [5] Cohen Priva (2015). *Informativity affects consonant duration and deletion rates*. [6] Thompson & Mulac (1991). *The discourse conditions for the use of the complementizer that in conversational English*. [7] Soldaini et al. (2024). *Dolma: an Open Corpus of Three Trillion Tokens for Language Model Pretraining Research*. [8] Honnibal & Montani (2017). *spaCy 2: Natural language understanding with Bloom embeddings, convolutional neural networks, and incremental parsing*.